“Secure Image sharing using Visual Cryptography”

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MAY, 2025

DECLARATION

I "**Krishita Sanjay Choksi**" having Enrolment Number **"101CTBMCS2122005"** hereby declare that

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With Sincere Regards,

Krishita Sanjay Choksi

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ABSTRACT

The Secure Image Sharing Using Visual Cryptography (VC) solution provides protection for sensitive data like medical records and confidential images whilst also dealing with unauthorized access issues. The security approach using traditional encryption is inadequate because it requires high computation resources whilst also remain susceptible to attacks like MITM. An image gets successfully encrypted through VC because it gets split into several cryptographic shares (2–8) which by themselves reveal no information until users combine all the shares correctly to rebuild the original picture. The system operates through a combination of XOR-based algorithm, Bit Level Decomposition (BLD), Modular arithmetic, Pixel Expansion (PE) and CMYK decomposition to create image shares from binary grayscale and coloured images. The design supports adaptive image sharing between client-server networks containing 1 server and 3 clients. The system performance uses PSNR, SSIM, NCORR and hash matching as metrics to ensure both image integrity and quality. The system achieves computational efficiency alongside superior security as one of its main features. Real-time storage and scalable deployment take place on AWS EC2 instances that use Amazon S3 for cloud-based data accessibility. The system architecture provides a secure method for efficient and scalable image sharing which works well for critical domains in healthcare and law enforcement.

Keywords: Secure Image Sharing, Visual Cryptography, AWS EC2 instances, Shares, Client-Server architecture, Quality metrics

LIST OF ABBREVIATIONS

Abbreviation	Description
AES	Advanced Encryption Standard
AWS	Amazon Web Services
BLD	Bit Level Decomposition
CMYK	Cyan, Magenta, Yellow and Black
DCT	Discrete Cosine Transform
DWT	Discrete Wavelet Transform
EC2	Elastic Compute Cloud
GUI	Graphical User Interface
LSB	Least Significant Bit
MA	Modular Arithmetic
NCORR	Normalized cross-correlation
PE	Pixel Expansion
PSNR	Peak Signal-to-Noise Ratio
RSA	Rivest–Shamir–Adleman
SSIM	Structural similarity index measure
SSS	Shamir's Secret Sharing
S3	Simple Storage Service
VC	Visual Cryptography
XOR	Exclusive OR operation

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1. INTRODUCTION

Data sharing is the process of transferring data from one system, user or device to another for information exchange, communication or processing. Although it is vital in various contexts, basic data sharing without the application of security measures is usually insecure and unsafe. When information is disseminated in the open without any form of encryption, authentication, or access control, it is exposed to several threats. For instance, in the process of transmitting the data through a network that is not secure, the attackers may launch a man-in-the-middle attack and get to modify or even capture the data [1]. In such cases, there can be leakage of information, loss of important data, and erosion of confidence among users or systems.

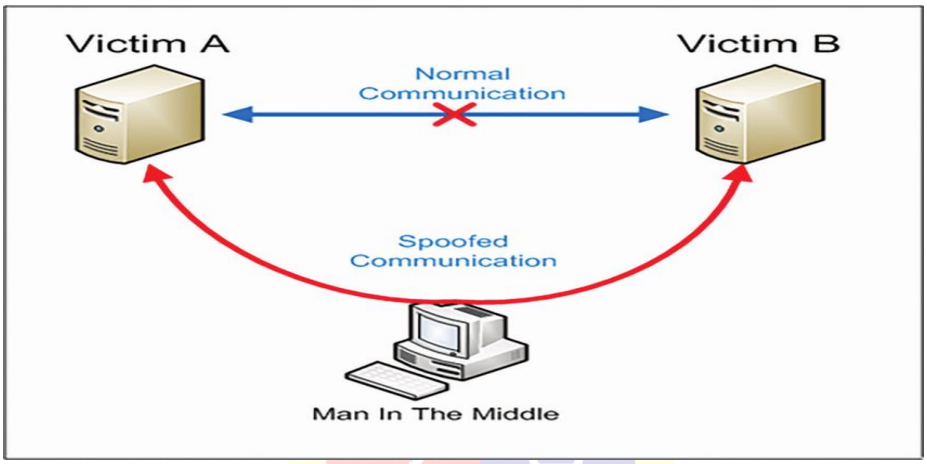


Figure 1.1 MITM attack [1]

Due to increased use of technology in communication, it has become important to safeguard images such as medical records, legal documents and personal pictures. As with most traditional encryption algorithms, this approach is ineffective when it comes to large and complex visual data. These encryption methods work very slowly as they need abundant computational resources whilst also remaining vulnerable to different attacks. Therefore, there is increasing pressure to use methods that are more efficient and lighter to implement with the aim of protecting image confidentiality.

This is where Secure Image Sharing Using Visual Cryptography (VC) provides a viable solution to this problem. The foundation of this concept lies in the classical secret sharing methods which deals with dividing the alphanumeric secrets into parts with threshold-based reconstruction systems [2] [3].The strength of this method is that it does not involve decryption of complex algorithms, which considerably reduce the number of computations. This approach is not only efficient, but also secure because the image is broken down into parts and cannot be retrieved unless the right shares are compiled.

The concept was generalized to visual data which was through developing VC for black-and-white images and share them into random-looking shares that can be combined to reveal the image directly to the human eye without complicated decryption [4]. For instance, a secure authentication QR code can

be divided into three shares and stored on different devices. It is only when at least two shares are aligned that the code becomes readable. But if the reconstructed image is displayed, for example, on a monitor under CCTV, it becomes insecure to shoulder-rushing attacks, replay, or replication [5]. For instance, an attacker can steal a reconstructed biometric or medical image through a surveillance feed and gain access to the system or personal information.

VC resolves multiple issues that occur in alternative approaches to secret image sharing. The basic visual processes in VC allow users to reconstruct images without complex computations so the system requires less computing power [6]. The reconstruction process with VC enables unencrypted visual recovery directly without causing reconstruction difficulties because it eliminates complex encryption schemes. Secret sharing with matrix-based reconstruction takes a lot of system resources because it needs multiple shares for image reconstruction whereas visual cryptography functions use minimal shares and therefore it requires fewer system resources [7].The secret image sharing approach is made simpler with VC which uses resource in required amount whilst delivering secure and transparent image sharing solutions [8].

Most often grayscale images are used in medical applications such as X-ray images where differences in intensity help in detecting an issue such as fracture or tumour. The RGB format is important in applications such as CCTV whereby full context is required as well as identification of the individuals in the picture. Binary images are used in simple and basic applications such as document scanning, OCR, barcode detection, etc. where only the black and white contrast is necessary. The main difference is in the depth of the information; binary images contain the least amount of information (1-bit), while grayscale images contain intensity details (8-bit), and colour images contain the most amount of information with full colour data (24-bit).

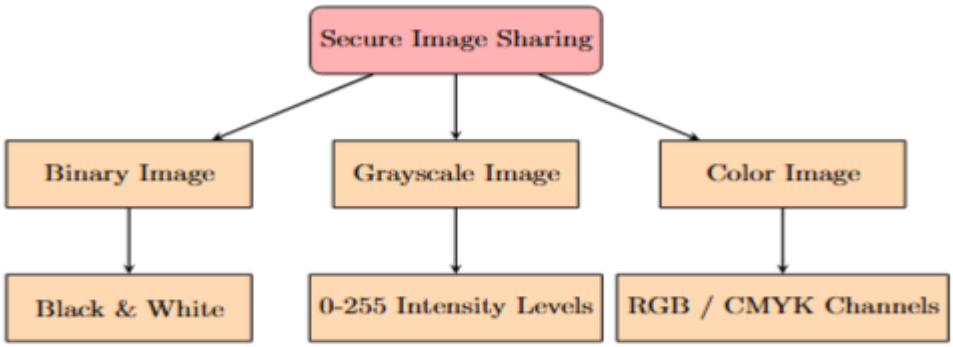


Figure 1.2. - Types of Images used [9]

The focus of Secure Image Sharing Using VC is on maintaining the confidentiality of sensitive visual information using advanced VC techniques. VC is a field of cryptography for encrypting visual data; the idea of which is to transmit images securely, breaking it into multiple cryptographic shares, so that no

individual share reveals any meaningful data. This makes it very resistant to interception and unauthorized access, for then it can only reconstruct the original image when the correct number of shares are combined.

The initial idea of VC is to disintegrate a secret image into multiple shares (between 2 to 8) and distribute them across multiple channels. The original image is only recreated by aligning the correct number of shares, which appear as random noise patterns each. By using this approach, the complex decryption algorithms are not needed, instead, we rely on the human eye or a simple computer algorithm to reconstruct the image if we have all shares.

Three types of images, namely binary, grayscale and colour images are generated cryptographic shares by this project using a combination of XOR based encryption, Bit Level Decomposition (BLD), CMYK decomposition , pixel expansion and modular arithmetic. These techniques break the images into shares that when combined, reconstitute the original visual information. This thesis also develops new cryptographic share creation methods based on modular arithmetic and CMYK halftone decomposition that have better security and usability in practice.

It is designed for flexibility of usage, and the shares are securely distributed to the client server system. User selects the required shares and the client gets the required shares and combines them to reconstruct the original image. This guarantees that the image is kept confidential while it is being transmitted, only the user has the required shares needed to decrypt it.

One of the important features of the system is that it is capable of giving performance evaluation using various metrics such as PSNR, SSIM, NCORR and encryption and decryption time, and this gives a deeper insight of the system's efficiency and quality of the reconstructed images. Also, the data integrity is ensured via hash matching and the cryptographic processes are monitored to verify the system performance with respect to the memory usage.

The system is deployed on AWS EC2 instances with Amazon S3 as a storage resource to ensure scalability, secure storage and real time access to the image shares.

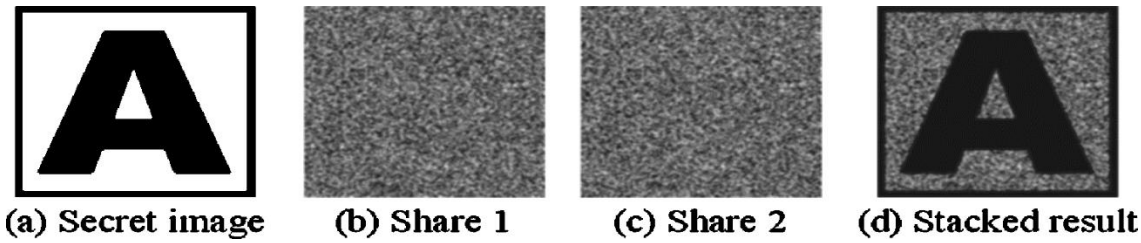


Figure 1.3 Visual Representation of VC working [10]

1.1 Motivation

Although, there are many methods of securing images like steganography, but steganography is not suitable for real time images or large images especially the ones that contain many pixilated images. For example, steganography involves hiding the secret image into the other cover image but the cover image can still be discerned or recovered if one scrutinizes it.

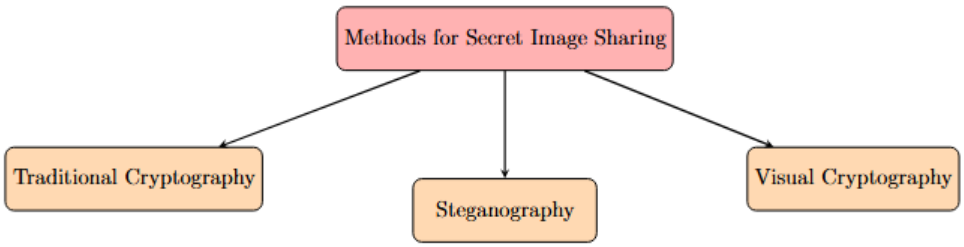


Figure 1.4 Types of methods used [9]

Visual Cryptography is different from the conventional cryptography in that it does not entail the use of complicated decryption or encryption keys. The security is present in the distribution of share images which, when taken individually, contain no significant information but when combined form a complete image. Visual cryptography provides protection against attacks because its unembedded data structure prevents detection methods and statistical analysis techniques. [11].

There are many scenarios where secure image sharing using Visual Cryptography is necessary, especially in contexts like digital forensics, medical and security system where the privacy of the images is of outmost importance. In these fields, it is crucial to prevent the images from being accessed, modified or exposed to threats by unauthorized personnel. For example, in the health care sector, patients' records must be shared with the other practitioners to improve efficiency yet the information should not be disclosed to unauthorized parties. The issues are solved in Visual Cryptography through the use of XOR encryption, Bit Level Decomposition and CMYK halftone decomposition to encrypt image data. These

methods ensure that only the rightful recipient with the right shares is able to access the full image and it's contents [12].

The integration of AWS Amazon Cloud and Linux makes the system more secure and scalable when it is incorporated further. AWS Cloud offers a secure environment in which the encrypted shares will be stored and accessed, so that the image data is safe not only during transmission but also while it is stored. The use of AWS EC2 instances and Amazon S3 as storage makes it easier to access image shares in a secure and real-time manner thereby making the reconstruction of images on call possible. It is also noteworthy that Linux serves as a highly efficient platform for executing encryption and decryption algorithms, thereby enhancing system stability during the processing and reconstruction of cryptographic shares. [13].

The logic of generating shares instead of sharing the images directly is very vital to maintain the aspect of the security in the system. In case one or several shares are intercepted or disclosed, it is still impossible to obtain any meaningful information from them since all the information is contained in the shares. This is much more secure compared to direct transfer of a full image since access is only granted to the recipient once all the shares have been received as planned. Also, the share-based systems are more flexible in controlling the flow of information in the organization since it is easier to limit the access to the shares than to limit users' access to the information. The method implements principles of threshold scheme of secret sharing by establishing a requirement for specific share combinations to regain the original secret. This technique provides extra security especially when many people are involved in the process of conveying the secret image [13].

Visual Cryptography is needed as a more effective and practical solution since it addresses issues that are not solved by traditional encryption and steganography. In the case of encryption, the image data can be retrieved if the key to the encryption is obtained while in the case of steganography the hidden content can be easily detected and extracted. In dividing images into shares, Visual Cryptography has the advantage that even when some of the shares are compromised, it is difficult to gain access to the original image. Therefore, VC provides another level of security that is not available in other methods especially where security of high levels of confidentiality is paramount [11].

There are many papers on secure image sharing and visual cryptography but there is a dearth of knowledge on the use of grayscale, binary and coloured images in the context of VC. Previous works are mainly based on monochromatic (black-and-white) or binary images while there is not much research done on how VC can be applied to coloured images. This is the research gap this project aims to fill, given that the current project focuses on the challenges of secure image sharing in such formats with an aim to enhance Visual Cryptography and extend its usefulness to real life complex images [14].This paper will discuss how grayscale, binary, and coloured images can be securely shared through VC and present solutions for industries that demand high security and privacy in image sharing.

1.2 Scope of the Project.

Secure Image Sharing Using Visual Cryptography project is for secure transmission of sensitive images using visual cryptography techniques. Through a Tkinter interface, the system allows the users to input three images: coloured, grayscale and binary. The sequence of the essential features and operations of the project are as follows:

1)Image Division into Shares Using Visual Cryptography:

After uploading the images into the system, the images go through XOR encryption, Bit-Level Decomposition, Pixel expansion algorithm, Modular Arithmetic algorithm and CMYK decomposition into multiple cryptographic shares. It is possible for the user to choose the number of shares, from 2 to 8. Each share by itself contains useless data, but a combination of the shares will reconstruct the original image [4].

2)Client-Server Model for Share Distribution:

The shares are distributed to different clients using a client server architecture. The original image is reconstructed only when the authorized recipients combine all the shares assigned to the respective clients [15].

3)Image Reconstruction and Quality Comparison:

Finally, the reconstructed image from the shares is used to calculate the PSNR, SSIM, NCORR, and Hash Match with the decrypted image. These comparisons help in maintaining and enhancing the quality of the image after encryption and decryption of images [16] [17].

4)Cloud-Based Deployment Using AWS:

The system also provides for cloud-based storage through AWS Console and S3 for secure image share storage and management. With this, encrypted images will be accessible seamlessly and you will be able to work remotely [18].

5)Real-Time Encryption and Decryption Performance Evaluation:

This project evaluates encryption and decryption system performance through testing of encryption speed and decryption speed as well as memory requirements and image reconstruction quality. These performance metrics are necessary to optimize the usage of the system in real world applications [19].

1.3 Organization of the Report

1. **Introduction:**

The visual cryptography techniques are used for secure image sharing in this project. It enables the user to upload coloured, grayscale or binary images using a Tkinter GUI and then use cryptographic methods to securely split and share images. The purpose is to guarantee the transmission security through allowing only authorized users to reconstruct the original image.

2. **Theoretical Background & Literature Survey:**

The research carries out the methodological evaluation of the papers focusing on the limitations, methods and the outcomes attained. Many of those methods fail to include the measures for preventing unauthorized access and does not have the specific security measures for the cloud infrastructure of the system or the mechanisms for restoring the high-quality images. This study does not consider implementation opportunities in relation to present-day technologies such as AI and blockchain systems.

3. **The Proposed Model and Implementation Methodology:**

The system generates cryptographic shares of images using Visual Studio while it relies on XOR encryption, BLD, PE, CMYK decomposition and MA algorithms. The share construction process in the system operates only during authorized client interactions to combine shares through client-server architecture using one server and three clients. The system performs evaluations through PSNR, SSIM, and NCORR performance metrics. Three different ec2 instances running on AWS amazon Linux machine performed the procedure through S3 buckets for both sharing duties and image handling.

4. **Theoretical & Empirical Result Analysis:**

This project evaluates encryption and decryption system performance through testing of encryption speed and decryption speed as well as memory requirements and image reconstruction quality. The reconstructed results were successful while showing minimal quality degradation through visual and numerical evaluation methods.

5. **Conclusion:**

A secure image sharing system is implemented through visual cryptography in this project. Images are transmitted via encrypted transmission with high efficiency and robust performance, and that makes the system suitable for secured communication.

6. **Future Work and Plan:**

This part details possible improvements for the system that include live system monitoring functions alongside machine learning analysis capabilities and multiple user access to dashboards. A proposed

Gantt chart presents the project development phases through a graphical timeline which includes design and development stages together with testing and reporting phases.

7. **References:**

The project references all research papers in addition to tools, APIs, articles, and libraries in this part. The formatting conventions for this document are either IEEE or APA depending on specific requirements.

2. Theoretical Background & Literature Survey

Secure image sharing is achieved by dividing visual data into multiple encrypted shares that can be reconstructed only by authorized users using visual cryptography (VC). These include converting sensitive documents to text and then sharing images, encrypting and storing in the cloud or using hybrid approaches that combine VC with other cryptographic algorithms. They also ensure the data's confidentiality, integrity and controlled access in cloud service environments. Nevertheless, existing methods are still subject to computational overheads, and apply to simple media data.

2.1 Secure Sharing of Data using variegated methods and concepts

Secret image sharing is possible through use of normal encryption and steganography among others. In classical cryptography there are other types of algorithms like AES, RSA which are employed to encrypt images before transmitting them. However, such methods are computationally intensive and highly dependent on key management; so if the keys are compromised then the data is compromised as well. On the other hand, Steganography is a technique where the image data is placed into an image, audio or video format which makes it very hard to detect whether or not there is anything hidden inside the cover data. Scientists have used steganography to hide images in JPEG or PNG format. However, it is less tolerant to compression, format changes, loss, and a range of steganalysis methods, which lowers the chances of successfully hiding the data.

The delivery of grayscale, coloured and binary images employ three techniques which include LSB, DCT and DWT, and palette-based techniques. LSB is a basic steganographic tool; however, security experts can easily recognize when images are altered or contain noise. Maintaining data robustness with DCT and DWT methods has its disadvantage because these methods consume a lot of computational power to execute which might lead to poor image quality. Palette-based colour schemes work well with indexed images but they do bring colour changes that are easily noticeable. Current steganalysis approaches effectively counter these techniques of data hiding which make data hiding even more challenging.

Technological revolutions in the field of cryptography have led to the creation of secure image sharing platforms ensuring confidentiality, access control and integrity of the visual content. Most importantly these systems are rather fitting for use in cloud environments where it is necessary to transfer and store information securely.

- **Visual Cryptography (VC):** A basic image security technique in which an image is split into multiple shares, each of which is individually meaningless. Partial access reveals no information unless the original image can be reconstructed only when the appropriate number of shares is combined.

- **Document Sharing with VC:** Some systems employ VC to turn sensitive documents into image-based shares and store them securely in the cloud. This type of text to image transformation is safe, even when storing content confidentiality, this approach is suitable.
- **Hybrid Cryptographic schemes:** The combination of virtualization cryptography and traditional security measures by using AES or RSA serves as an advanced encryption practice. These security schemes add higher levels of operations to systems which leads to increased computational work yet produce better security results.
- **Cloud-based VC scheme:** Encrypted image shares can be stored and managed in cloud environments using Cloud-Based VC Storage Platforms. The original image can be recovered only by authenticated users with the right shares and access rights, thus providing secure multi user access control.
- **Traditional Cryptography:** Digital files can be protected by traditional cryptographic methods through use of AES and RSA The encryption techniques convert whole image files into unrecognizable ciphertext through a process which requires the proper decryption key for decryption to be successful. High security stands alongside key management overhead and computation overhead and extra encryption infrastructure as features of these crypto methods.

The major drawbacks of these tools involve:

1. Limited to cases where the content can be structured.
2. The lack of an optimization feature conflicts with performance enhancements through computational efficiency and head optimization.
3. Hybrid schemes created additional complexity for the system.

Our project creates an easy-to-use open-source platform that protects sensitive images and It allows users to:

- The system divides input pictures into different shares according to algorithm.
- It allows reconstruction of images only when multiple users confirm their access permissions.
- The system uses less processing power to protect sensitive data properly.

2.2 Comparative Analysis of Existing Schemes.

This section evaluates papers based on Secure Image sharing using different methods. Each paper possesses its own practicality level alongside unique methodologies and techniques to secretly transfer images without revealing their true nature and original input image.

Table 2,1 Comparative Analysis of Research Papers

Citation Number	Methodology	Technique used	Limitations	Results
[20]	A secure scheme for transferring documents using VC was found for cloud specifically.	Converted document files into text, then into image shares using VC and stored them in cloud.	The system handles mainly text documents yet needs more development to handle advanced image information.	The system established efficient data management and retrieval practices while maintaining complete protection of data confidentiality together with integrity security.
[21]	Added a new way to share secrets through visual cryptography and its improvements.	Divided picture files into parts to ensure that users can rebuild the image files only when they have proper approval.	The system has security weaknesses when unauthorized users acquire enough shares yet it lacks features for cloud environments.	The system helped users set access rules for their secret images based on their account roles.
[20]	Created a safe way to share images through visual cryptography on cloud systems.	The system transformed images into encrypted files then uploaded them to the cloud where only approved users could view the real images.	The work did not explain how its solution affected computer performance and storage space requirements.	Offered a safe way for users to save and share images in cloud storage systems.
[22]	The author combined visual cryptography with other security systems to	Visual cryptography works better when combined with extra protection	Complexity is increased due to integration of multiple cryptographic methods	Better protection for cloud data sharing activities.

	create a new data sharing method for cloud environments.	methods to secure transmitted data.		

2.3 Research Findings and Research Gap

1. **Large Scale Data:** Many of the existing visual cryptography techniques are not optimized for large scale image dataset or multi user environment. The design of scalable and efficient solutions to deal with the plethora of data and large volumes of high- resolution images without hindering the security or the system performance is a key research gap.

2. **Risks of Unauthorized Access:** Although [21] recognizes that there is a risk of unauthorized users being able to access sufficient shares, it does not explore the possibility of adaptive mechanisms to detect or prevent such access in real time. In environments with limited trust, there is a need for a robust mechanism that allows for the combination of visual cryptography with anomaly detection systems to stop unauthorized access.

3. **Specific Cloud Security Issues:** The works ([21], [22]) present secure image sharing in cloud, but they do not address the security issues related to cloud such as the vulnerabilities in storage, data migration hazards, and dependency on the cloud provider. More work needs to be done to a) explore more specifically how visual cryptography can be adapted to overcome these cloud particular challenges

4. **Integration with emerging technologies:** Having been extensively studied, the study of the visual cryptography has not been extended to the integration with the recent advancement in technologies including the blockchain, edge computing, and AI based anomaly detection.

5. **High-quality image recovery:** Visual cryptography often leads to slight degradation of the original image, especially when applied to more complex images, but this degradation is lossless or high quality. The problem is to develop lossless or near lossless image recovery techniques.

3. The Proposed Model and Implementation Methodology

3.1 The Problem Statement

Within the context of the existing computerized environment, the protection of visual data and their transmission through insecure channels, and more specifically through the cloud, is still an open problem. Unfortunately, conventional encryption techniques still pose challenges of key management and become virtually useless if the encryption key is intercepted by an unauthorized person. In addition, with the development of the cloud platforms, the problem of sharing images while maintaining confidentiality and providing access to them only to the authorized persons, excluding the use of cryptographic keys, has become relevant.

Visual cryptography is a good solution for this problem since it does not require key exchange procedures. However, existing techniques have disadvantages that include the following:

1. Limited Domain of Application: Most of the current systems are designed to work with limited type and size of images and are not capable of handling high resolution or complicated images across different domains [15].
2. Access Control Limitations: The threshold-based schemes are also not very safe because there is always a possibility of unauthorized reconstruction of the images if the share access is not properly controlled or supervised [13].
3. Some frameworks do not address the practical concerns of cloud computing such as user authentication, different levels of privileges, storage, and data migration issues [18].
4. High Computational Overhead: While implementing VC with traditional encryption leads to improved security, the computation time and the memory usage are considerably high, making it unsuitable for real-time applications [6] [13].
5. Scalability: Most of the existing systems are not designed for multi-user or distributed systems, thus they do not possess the lightweight and modularity that is required for scalability in environments such as edge or cloud computing [17].

Therefore, the main problem this project tries to solve is:

"To design an effective and secure visual cryptography-based image sharing mechanism which provides restricted access, privacy and integrity of the shared image in cloud environment with less dependence on the key-based cryptography and computational cost."

3.2 System Model / Architecture Overview

The proposed system follows a client server architecture suitably developed for sharing grayscale, binary and coloured images using five different algorithms available in visual cryptography. Such architecture covers the main modules for generation of shares, the distribution of shares with restricted access, reconstruction of shares, and performance assessment.

1) First Module

This module focuses on the selection of images from publicly available online sources and the generation of shares for the selected images. The system starts with Tkinter GUI to help the user select only three input images.

2)Second Module

These selected images are encrypted using encrypt() function out of following encryption methods:

XOR

This encryption technique is quite similar to encrypting a number with the help of XORing with a secret number. This is made to work for every pixel value in an image in order to extend the idea. The original data is then combined with k share of images of the same size using the XOR operation where k>1. This secret image is then again combined through the Exclusive OR operation with the k share images to retrieve back the original image. Hence this is an (N, N) Sharing scheme where all the k share images as well as the secret image is required to decrypt the image. This is because, in order to reveal the original picture and obtain the secret picture, secret image is necessary. If one share image is missing, then it would take 2^(m*n) states to get the final image where each state is equiprobable, it becomes difficult to decrypt the image of size m * n which is the size of the original image.

Modular Arithmetic

This encryption technique works on the basis of modular arithmetic and the cyclic nature that is associated with it. Since the value of each pixel is an integer in the range of 0 to 255, the modulo operation by 256 is applicable. The original image is added with k share images (where k>1) of the same size and then the result is taken to modulo 256 to obtain the final secret image. To decrypt, each k share image is subtracted from the secret image by taking the result modulo 256 to obtain the final decrypted image. This algorithm is an (N, N) Sharing scheme, and as such all k share images along with the secret image is needed in order to decrypt the image. If one share image is missing, the number of states is 256^(m*n) to get the final image in which every state is equally likely, making it difficult to decrypt.

Bit Level Decomposition

This algorithm extends the PE technique to grayscale images by first decomposing the input into multiple binary layers, upon which the PE process is applied. The two obtained binary images are then integrated to generate two grayscale shares. The same procedure is followed in the decryption of the image. Since the pixel value in the image ranges from 0-255, the original image is separated into 8 binary images. In the case of the binary image, value is the same as what value is present in the pixel in that bit. For instance, (i, j)th pixel in kth binary image will be the kth bit of the (i, j) pixel value in the original image. We have used two methods of decryption because of the Pixel-Expansion algorithm: In the overlap phase, the decrypted image initially appears at twice the original size and is subsequently resized to half for proper alignment. During extraction, a pixel is considered black only if all its corresponding subpixels are black; otherwise, it is classified as white.

CMYK Decomposition

Initially, visual cryptography was reported for only binary images but recent contributions have been made towards its uses in grayscale and colour image. One of the methods is when the colour image is segmented into CMYK. Based on the first three images, we apply halftone conversions on the three monotone images. We then apply the nearest neighbour interpolation to come up with the three shares. We later combine the three to obtain the output image file in the form of a share.

Pixel Expansion

In this algorithm, the pixel is divided into subpixels and the whole image is divided into two shares; the first share is overlapped on the second share to obtain the decrypted image. All the subpixels in each collection will be in an equal proportion of black and white subpixels. The colours of the sub-pixels are divided in such a way that even when the two shares are overlapped, the previously black coloured pixel will have all the sub pixels black and the previously white coloured pixel will have half of the sub pixels black. We have used the four subpixels for each pixel implementation algorithm. This would make the share images to be twice the size of the actual image size that is displayed on the website. As there is 4C2 options for selecting subpixels in each pixel of the share randomly, decryption using a single share is quite impossible and requires 6^(m*n) states where m*n is the size of the original image for a brute force attack to decrypt the secret from a single share. We have tried two methods to decrypt : In the overlap phase, the decrypted image initially appears at twice the original size and is subsequently resized to half for proper alignment. During extraction, a pixel is considered black only if all its corresponding subpixels are black; otherwise, it is classified as white.

3)Third Module

After the encryption of selected images using a particular method and for a particular type of image the model is set to produce between 2 to 8 cryptographic shares for each image.

The shares are saved as .png,.jpg or .jpeg file respectively as well as are shared automatically to:

Client 1 (first n1 shares),

Client 2 (next n2 shares),

Client 3 (remaining shares).

This distribution structure enables threshold forms of access control to combat any possibility of reconstruction by unauthorized individuals.

4)Fourth Module

Forwarding & Intermediate Aggregation (Client 2)

It is noted that Client 2 retrieves shares from both Client 1 as well as its own directory.

These are merged and forwarded to Client 3 in a structured directory (combined/) for decryption.

This layer mimics the delivery of share in real-life settings where different parties are allowed partial access.

5)Fifth Module

Share Combination, Decryption & Evaluation (Client 3)

The shares are then loaded and decrypted with the help of the decrypt() function from the decryption module file, which uses either XOR, MA, BLD, PE or CMYK Halftone Decomposition algorithm according to the selected encryption method.

The reconstructed image is saved and then compared with the original using:

1. PSNR, NCORR, SSIM, Hash integrity metrics
2. CPU usage during encryption and decryption
3. Memory usage during encryption and decryption
4. Encryption and decryption time taken to complete the process
5. Bars and Plots to visually see the difference between 2 images.

6)Final Module

Repeating the process on AWS console's Amazon Linux using 3 instances and s3 bucket for downloading and copying shares as well as images for enhanced security and authorized access.

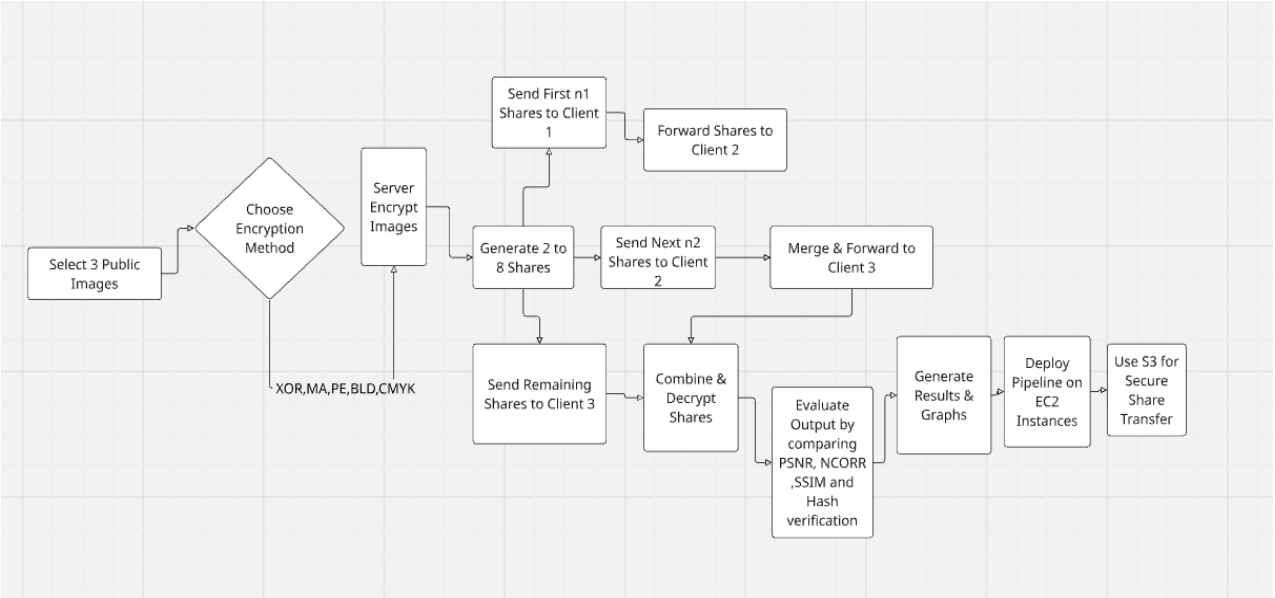


Figure 3.1- System Flowchart [9]

4. Theoretical and Empirical Analysis

4.1 Experimental Setup

4.1.1 Libraries to be installed

Table 4.1 Libraries used

os	For creating directories, managing file paths and for navigation to the folder where shares/images are to be saved and accessed.
time	Performance evaluation of the system relies on recording time measurements from both encryption and decryption processes.
tracemalloc	Monitors the amount of memory used in encrypting and decrypting of images in order to deduce the utilization of the system resources.
numpy	Converts the images to and from arrays; performs the matrix operations needed for XOR and other pixel-level processing and modular arithmetic.
Psutil	Collects the performance data of the system, at the process of cryptographic operations, including the CPU and memory usage.
Pillow	The library manages image input and output processing between different data formats (PNG, JPEG) and performs colour conversion operations (RGB to grayscale).
Matplotlib	Present the results in form of plots and graphs such as bar graphs of PSNR, SSIM, and encryption time.
Seaborn	Implements functions to improve the appearance of the plots, especially for comparison of heatmaps or bar charts.
Skimage	This paper presents the structural_similarity function (SSIM) that can be used to evaluate structural integrity of decrypted images.

4.1.2 Dataset

1. Grayscale Input Images:

Grayscale images are required in this project because they give the right level of details while being easy to process. As they have pixel values ranging from 0 to 255, they store the brightness levels without colour, thus making their use appropriate for visual cryptography techniques such as XOR, bit-level decomposition, and modular arithmetic. Thus, it is possible to generate shares and reconstruct the original image with high accuracy and at the same time they contain more details than binary images due to their simpler structure.

Figure 4.1.1 Grayscale Pebble

565 x 849

Figure 4.1.2 Grayscale Girl

1120 x 1120

Figure 4.1.3- Grayscale Dog

500 x 350

Figure 4.1- Grayscale Images

2. Binary Input Images:

Binary images are an essential part of this project because they are the simplest form of images, which contain only two values – 0 and 1. This makes them highly appropriate for fundamental operations like XOR as well as simple arithmetic operations such as MOD. While not being very detailed in terms of image display, binary images are easy to process and take up a very small amount of space, thus they are effective in terms of share generation. This is because contrasting colours are easier to differentiate and this enhances the effectiveness of the visual reconstruction particularly where detail is not very paramount.

Figure 4.2.1 Binary Rings

228 x 221

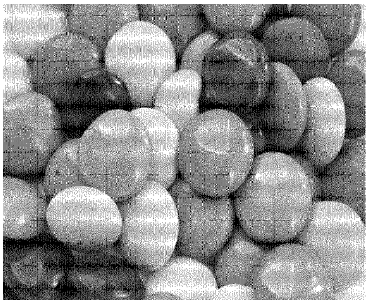


Figure 4.2.2 Binary Pebble

565 x 849

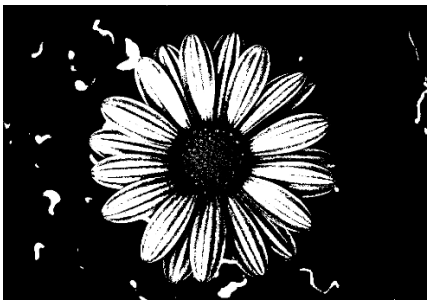


Figure 4.2.3 Binary Flower

1000 x 688

Figure 4.2 Binary Images

3.Coloured Input Images:

In this project colour images hold great importance because they show detailed visual data using multiple channels which normally use the RGB format with Red, Green and Blue components. The three colour components in RGB format occupy dedicated spaces in every pixel which generates complex visual information about an image. Coloured images prove most suitable for applications which demand complete visual context such as object detection systems or security monitoring or image identification tasks. The combination of increased storage requirements and processing power consumption by coloured images remains worthwhile since they supply critical colour information which leads to precise analysis in advanced tasks.

Figure 4.3.1 Colour Blast

1170 x 700

Figure 4.3.2 Colour Bomb

643 x 360

Figure 4.3.3 Colour Pebble

565 x 849

Figure 4.3 Coloured Images

4.2 Evaluation Metrics

4.2.1 Parameters Used

1. PSNR

The program uses the PSNR formula to evaluate the quality of contrast image by comparing it to its original version.

A measurement indicating the quality level of contrast images based on their noise content compared to original content. The quality assessment of images is better when PSNR reaches higher values.

PSNR	$10\log_{10}\left(\frac{MAX_I^2}{MSE}\right)$ MAX_I: The maximum possible pixel value in the image MSE: The mean squared error between the two images

Figure 2.4 PSNR [23]

MSE computes the average squared deviations which exist between pixel values of original and contrast images.

Where PIXELMAX is the maximum pixel value (255 for 8-bit images), and the higher the PSNR, the better quality of the contrast image relative to original image.

2. SSIM

The SSIM serves as a perceptual metric to evaluate image quality deformation which results from processing such as compression and encryption/decryption steps. The metric was created to assess transformations in structural composition together with brightness and contrast levels.

SSIM	$\frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$ μ_x and μ_y: The mean brightness values of the two images σ_x^2 and σ_y^2: The variance of each image σ_{xy}: The covariance between the two images C_1 and C_2: Constants for stability

Figure 4.5 SSIM [23]

3. NCORR

normxcorr2D() function utilizes an NCC formula designed for localization to measure template-image similarities across all positions within the image.

$NCC = (\sum(I(i, j) - \mu I)(J(i, j) - \mu J)) / (\sqrt{(\sum(I(i, j) - \mu I)^2)} * \sqrt{(\sum(J(i, j) - \mu J)^2)})$ [24]

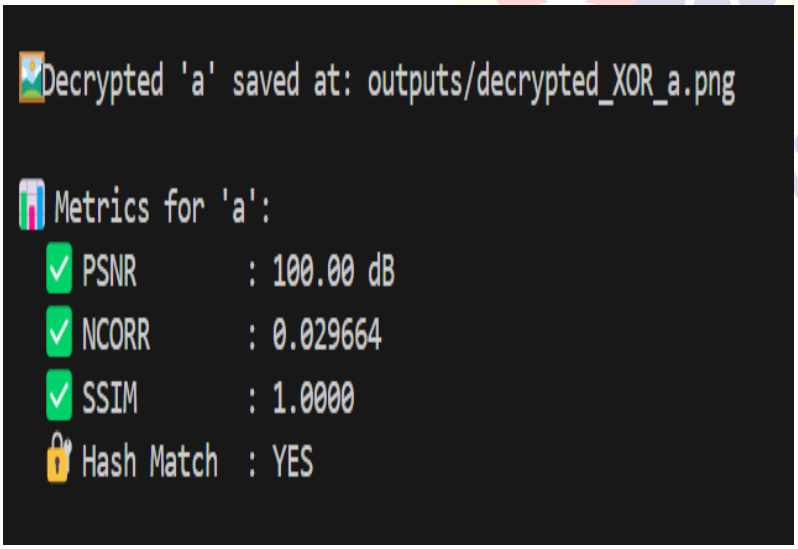
- μI is the mean intensity of image I [24]
- μJ is the mean intensity of image J [24]
- $\sum$ represents the sum over all pixels (i, j) [24]

4. Hash Verification

The serialization of the image begins with transforming it into a raw byte array for processing. A SHA-256 cryptographic hash function creates a unique 256-bit digest through its application. The generated hash functions as a secure identification tool that represents the image content to check its integrity and spot deepfake alterations.

4.2.2 Grayscale Pebble evaluation

Grayscale Pebble evaluation for XOR method



Grayscale Pebble for BLD method

```
🔍 Processing image: a
🔐 Total shares used: 6
 - shared/client3\a\combined\share_1.png
 - shared/client3\a\combined\share_2.png
 - shared/client3\a\combined\share_3.png
 - shared/client3\a\share_1.png
 - shared/client3\a\share_2.png
 - shared/client3\a\share_3.png
✅ Decrypted images saved as:
 - outputs/decrypted_BLD_overlap_a.png
 - outputs/decrypted_BLD_extraction_a.png

📊 --- Evaluation (Extraction Image) ---
📈 PSNR: 100.00 dB
🔎 NCORR: 0.0297
🔐 Hash Match: YES ✅
```

Grayscale Pebble for Modular Arithmetic method

```
🔐 Total shares used for decryption of 'a': 8
 - shared/client3/a/combined\share_1.png
 - shared/client3/a/combined\share_2.png
 - shared/client3/a/combined\share_3.png
 - shared/client3/a\share_1.png
 - shared/client3/a\share_2.png
 - shared/client3/a\share_3.png
 - shared/client3/a\share_4.png
 - shared/client3/a\share_5.png
✅ Decrypted image 'a' saved at: outputs/decrypted_MA_a.png

📊 --- Evaluation for 'a' ---
📈 PSNR: 100.00 dB
🔎 NCORR: 1.000000
📊 SSIM: 1.0000
🔐 Hash Match: YES ✅
```

Table 4.2 Evaluation Metrics for Grayscale Pebble

Evaluation Metrics	XOR	Modular Arithmetic	Bit Level Decomposition
PSNR	100.00 dB	100.00 dB	100.00 dB
NCORR	0.029644	1.00	0.0297
SSIM	1.00	1.00	1.00
HASH MATCH	YES	YES	YES

4.2.2 Binary Pebble Evaluation

Binary Pebble evaluation using XOR method

```
📊 --- Evaluation ---
📈 PSNR: 100.00 dB
🔎 NCORR: 1.000000
📊 SSIM: 1.0000
🔐 Hash Match: YES ✅
```

Binary Pebble for PE method

```
Enter image ID (e.g., 'Input'): Input
🔐 Loading shares...
✅ Decrypted and saved overlapped and extracted images.

📊 --- Evaluation ---
📈 PSNR: 100.00 dB
🔎 NCORR: 1.000000
📊 SSIM: 1.0000
🔐 Hash Match: YES ✅
```

Table 4.3 Evaluation metrics for Binary Pebble

Evaluation Metrics	XOR	Pixel Expansion
PSNR	100.00 dB	100.00 dB
NCORR	1.00	1.00
SSIM	1.00	1.00
HASH MATCH	YES	YES

4.2.3 Coloured Pebble Evaluation

Coloured Pebble evaluation using XOR method

```
🔐 Total shares used for decryption: 8
 - shared/client3/col/combined\XOR_share_1.png
 - shared/client3/col/combined\XOR_share_2.png
 - shared/client3/col/combined\XOR_share_3.png
 - shared/client3/col\XOR_share_1.png
 - shared/client3/col\XOR_share_2.png
 - shared/client3/col\XOR_share_3.png
 - shared/client3/col\XOR_share_4.png
 - shared/client3/col\XOR_share_5.png

✅ Decrypted image saved at: outputs/decrypted_XOR_col.jpeg

📊 --- Evaluation ---
📈 PSNR: 100.00 dB
🔎 NCORR: 1.000000
📊 SSIM: 1.0000
🔐 Hash Match: YES ✅
```

Coloured Pebble for CMYK halftone decomposition

```
✅ Decrypted image saved at: outputs/decrypted_CMYK_coloured.png
📐 Input Image Size: 849 x 565 pixels
📐 Decrypted Image Size: 849 x 565 pixels

📊 --- Evaluation ---
📈 PSNR: 32.82 dB
🔎 NCORR: -0.056892
📊 SSIM: 0.8028
🔐 Hash Match: NO ❌
```

Coloured Pebble MA method

```
PS D:\ImageShareSystem> python client3color.py
Enter image ID (e.g. 'Input'): coloured

🔐 Total shares used for decryption: 8
 - shared/client3/coloured\combined\MA_share_1.png
 - shared/client3/coloured\combined\MA_share_2.png
 - shared/client3/coloured\combined\MA_share_3.png
 - shared/client3/coloured\MA_share_1.png
 - shared/client3/coloured\MA_share_2.png
 - shared/client3/coloured\MA_share_3.png
 - shared/client3/coloured\MA_share_4.png
 - shared/client3/coloured\MA_share_5.png

✅ Decrypted image saved at: outputs/decrypted_MA_coloured.png

📊 --- Evaluation ---
📈 PSNR: 100.00 dB
🔎 NCORR: 0.016943
📊 SSIM: 1.0000
🔐 Hash Match: YES ✅
```

Table 4.4 Evaluation Metrics of Coloured Pebble

Evaluation Metrics	XOR	Modular Arithmetic	CMYK Halftone
PSNR	100.00 dB	100.00 dB	32.82 dB
NCORR	1.00	0.016943	-0.056892
SSIM	1.00	1.00	0.8028
HASH MATCH	YES	YES	NO

4.3 Test scenario/ Use case

The test application used a Tkinter-based GUI to choose from grayscale, binary and coloured images followed by encryption and decryption processes according to pixel properties. The system operates with five encryption methods including Pixel-wise Secure Masking through XOR, Modular Arithmetic and grayscale image decryption with Bit-Level Decomposition, binary subpixel division using Pixel Expansion and colour image encryption through CMYK Decomposition with Halftone conversion. The

distribution of 2 to 8 cryptographic shares for each image is managed by three clients to achieve threshold-access control. Client 3 receives aggregated shares from Client 1 and 2 through Client 2 before executing a matched algorithm for final decryption. The verification process for decryption accuracy involved PSNR, SSIM tests, NCORR calculations and hash integrity checks while system measurements included CPU and memory usage during encryption and decryption times. AWS Amazon Linux hosts a complete duplicate of the workflow employing S3 buckets along with three EC2 instances to demonstrate authorized real-time access within distributed systems that offer secure scalability.

4.4 Theoretical Analysis

Parameters used for theoretical analysis:

1)Encryption Simulated Time: The encryption speed measures the time needed to generate encrypted share images from an original datum. Real-time secure applications find support through encryption processes that occur at increased speeds. During testing the project evaluated encryption speed for grayscale as well as binary and colour images.

2)Peak Memory During Encryption: The measurement displays the largest amount of memory that was utilized during image encryption. The chosen encryption methods effectively function on systems with minimal resources because they demonstrate low memory usage during encryption operations.

3)CPU Usage During Encryption: The measurement reflects the processor use during encryption operations.

4)Decryption Time: The process of reconstructing the source image from distributed shares takes the measure called decryption time. The method achieves speedy and safe retrieval of data due to its shortened times.

5)Peak Memory During Decryption: The system monitors the maximum RAM consumption that occurs during image decoding.

6)CPU Usage During Decryption: The displayed measurement represents the processing demand on the processor while it reconstructs images. The performance of the CPU enables painless decryption of decrypted files even when processing multiple data types simultaneously.

For Grayscale Pebble

Table 4.5 Theoretical Analysis of Grayscale Pebble

Metrics	Encryption time	Decryption time	CPU usage during encryption	CPU usage during decryption	Peak memory during encryption	Peak memory during decryption
XOR	0.0109 seconds	0.0836 seconds	3.60 %	5.40 %	0.036 MB	7.6514 MB
MA	0.0110 seconds	0.0994 seconds	2.30%	6.10%	143.53 MB	148.70 MB
BLD	74.1880 seconds	76.3203 seconds	1.10%	8.60%	1054.00 MB	1054.00 MB

For Binary Pebble

Table 4.6 Theoretical Analysis of Binary Pebble

Metrics	Encryption time	Decryption time	CPU usage during encryption	CPU usage during decryption	Peak memory during encryption	Peak memory during decryption
XOR	1.1074 seconds	1.0366 seconds	8.20 %	7.60 %	4.4475 MB	7.3245 MB
Pixel Expansion	1.0001 seconds	1.0004 seconds	1.00%	1.00%	4.12 MB	4.18 MB

For Coloured Pebble

Table 4.7 Theoretical Analysis of Coloured Pebble

Metrics	Encryption time	Decryption time	CPU usage during encryption	CPU usage during decryption	Peak memory during encryption	Peak memory during decryption
XOR	0.2012 seconds	0.2223 seconds	2.20 %	4.75 %	143.60 MB	169.23 MB
MA	0.1422 seconds	0.0598 seconds	6.50%	6.00%	143.70 MB	158.420 MB
CMYK halftone	0.2016 seconds	0.0510 seconds	9.40%	1.25%	138.10 MB	138.90 MB

4.5 Implementation Result Analysis

Visualization of metrics for Binary Pebble using Pixel Expansion Method

1)There are two methods used for decryption: Overlap (the output image must be resized to half its original size because it is twice as large) and Extraction (where white pixels are extracted from a black area when all their sub pixels are black). The resized components were reduced to half their original size

for a match with the original picture. Pixels containing complete black subpixels are classified as black while the rest become white in this method.

2) The pixel expansion method establishes complete binary image reconstruction by exactly replicating the original content among the original, decrypted and overlap visuals. The method proves lossless as the decrypted image achieves a PSNR of 100 dB, NCORR of 1.0, and SSIM of 1.0 which makes it identical to the original image.

3) The system exhibits low CPU utilization below 1% while maintaining around 4.1 MB of memory usage throughout encryption and decryption processes. The method demonstrates operational efficiency which makes it appropriate for use in lightweight application scenarios.

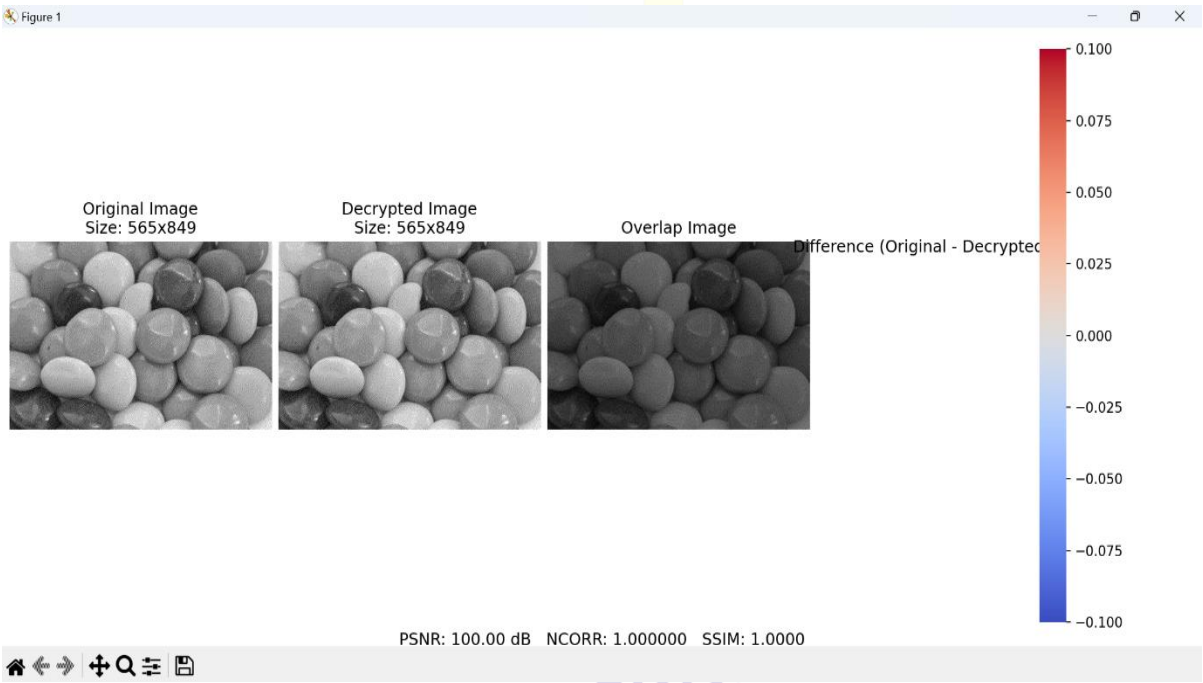


Figure 4.6 Binary Pebble Image Visualization

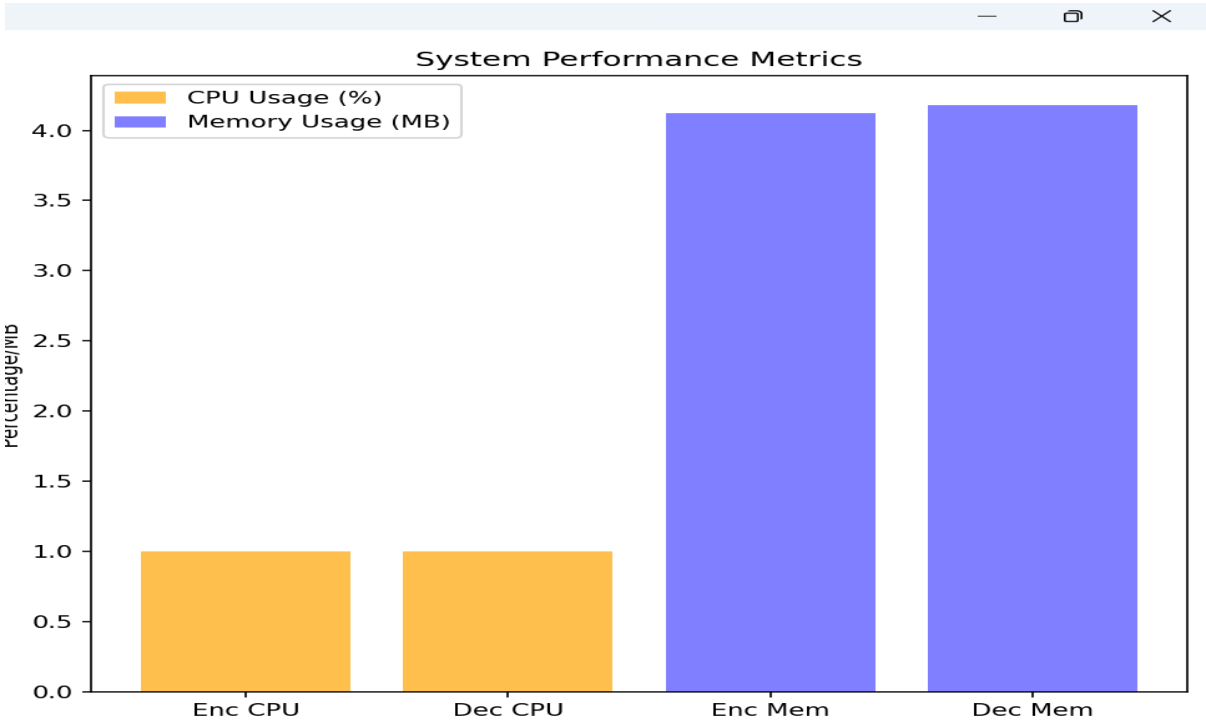


Figure 4.7 Binary Pebble Image Performance Metrics

Visualization of metrics for Grayscale Pebble using XOR Method

1)The encryption process bears resemblance to number encryption through XOR combined with secret numbers. The method extends its principle to perform encryption on pixel values throughout an entire image. A secret image results from XORing the original image with k share images of the same size to form a secret image.

2) A grayscale image shows an XOR encryption and decryption process in the initial presentation. A pixel difference heatmap verifies the exact match of the original and decrypted images which measure 565x849. These images appear to be completely identical. Quality measurements point to near-perfect reconstruction because PSNR reaches 100.00 dB while SSIM stays at 1.0000 and NCORR reaches only 0.0297.

3) A bar chart displays these metrics at the second image. The maximum PSNR value and perfect SSIM rating combined with minimal NCORR demonstrate the successful implementation of XOR for lossless encryption during image communication.

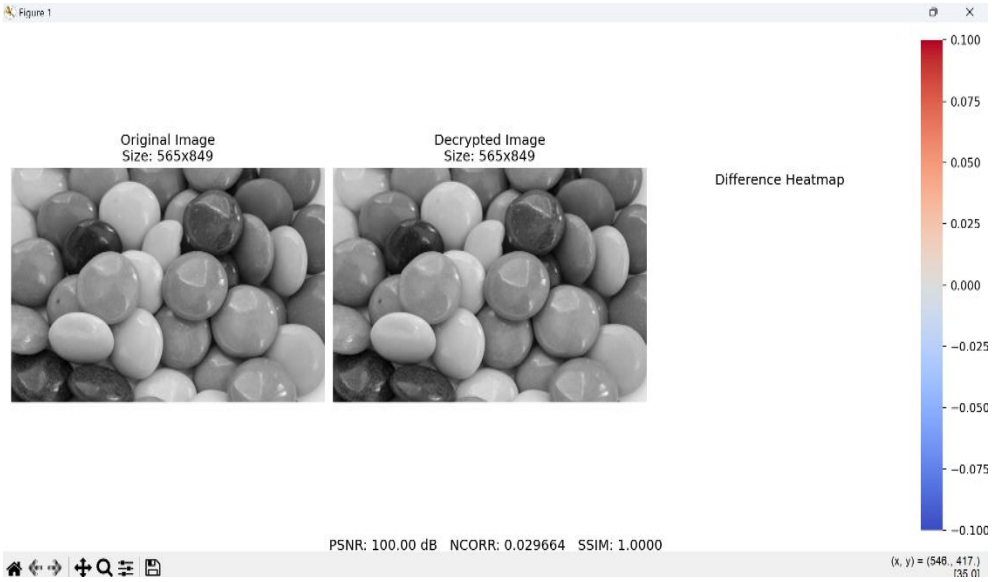


Figure 4.8 Grayscale Pebble Image Visualization

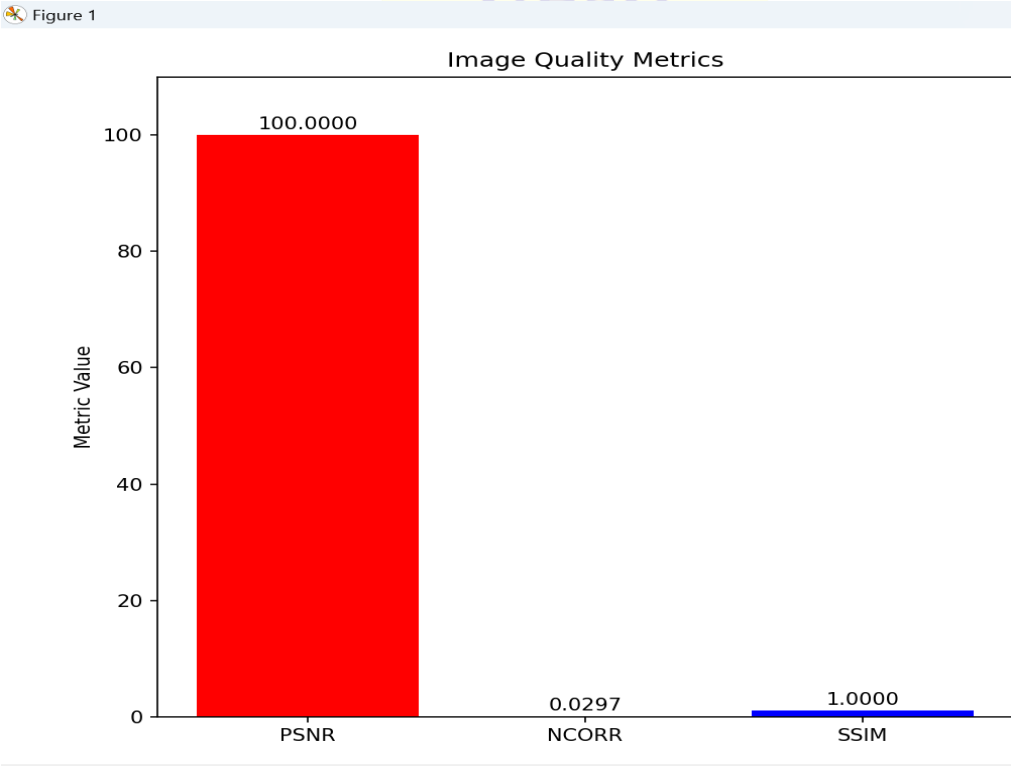


Figure 4.9 Grayscale Pebble Image Quality Metrics

Visualization of metrics for Coloured Pebble using MA Algorithm

1)This algorithm is an (N, N) Sharing scheme, and as such all k share images along with the secret image is needed in order to decrypt the image. If one share image is missing, the number of states is 256^(m*n) to get the final image in which every state is equally likely, making it difficult to decrypt.

2) The first figure presents the original image and its decrypted version as adjacent 565×849 pixel-sized images. Both pictures show no visible differences and the difference heatmap match with the original minus the decrypted version. A perfect image reconstruction emerges from PSNR at 100.00 dB while SSIM reaches 1.0000. The encryption-decryption process introduces minimal noise or distortion because

the Normalized Cross-Correlation (NCORR) value reaches only 0.016943. All tested metrics provide evidence that confirms the cryptographic method demonstrates accurate and dependable performance.

3) A bar chart in the second figure displays the system resource consumption for both encryption and decryption operations. The encryption process uses minimal processing power because the CPU usage remains between 2% to 5% indicating systems operate efficiently with this method. Both encryption operations require 144 MB memory resources whereas decryption operations need 159 MB memory consumption. Memory use surged during runtime based on potential reasons such as image processing libraries or buffer allocation processes. Optimization of this method should be done before deploying it to systems with limited memory resources.

Figure 4.10 Coloured Pebble Image Visualization

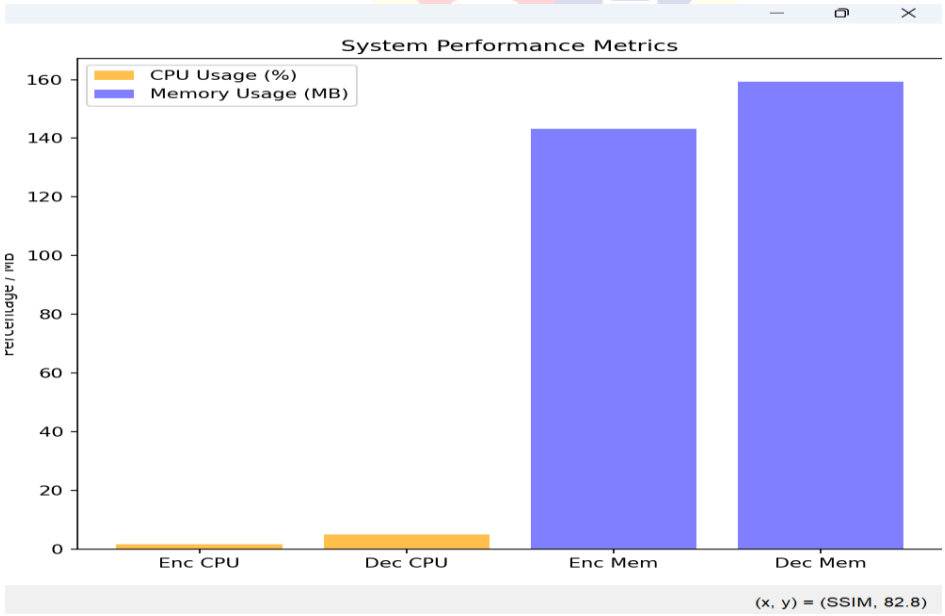


Figure 4.11 Coloured Pebble Image Performance Metrics

4.6 Comparative Analysis

Here we would be doing comparative analysis of evaluation metrics and theoretical analysis for Grayscale Pebble (via XOR) in Visual Studio Code and AWS Amazon Linux:

1)Encryption Time: VS Code took only 0.0109 seconds for encryption processing while AWS Linux needed 0.102 seconds to complete the same operation. Diverse system configurations may explain the performance discrepancy between VS Code running locally versus AWS Linux instances which may experience additional latency impacts from share virtualized CPU functions.

2)Decryption Time: The decryption operation on AWS Linux required 0.0494 seconds to complete yet VS Code needed 0.0836 seconds. The analysis indicates possible advantages for the decryption algorithm due to Linux environment's optimized I/O handling and memory access patterns despite initial delays during encryption.

3)CPU Usage During Encryption: The CPU encryption workload in VS Code reached only 3.60% while AWS Linux used its entire CPU capacity (100%) during same encryption process. A surge of CPU utilization on AWS reveals that a process monopolized an entire virtual core probably because of resource partitioning and minimal processor throttling and cloud environment dynamics which could affect short-term performance assessment.

4) CPU Usage During Decryption: The decryption process on AWS reached 100% CPU usage yet VS Code only used 5.40% during this operation. The allocation of a single thread or vCPU by AWS reached its maximum during the brief decryption process because VS Code could distribute the workload better across available processing cores.

5) Peak Memory During Encryption: The memory consumption levels during encryption remained steady at 0.036 MB for VS Code and 0.034 MB for AWS Linux. The minor variation in memory usage demonstrates that XOR logic maintains its memory efficiency in all circumstances.

6)Peak Memory During Decryption: The implementation of decryption yielded nearly equal peak memory usage statistics between VS Code at 7.6514 MB and AWS at 7.6480 MB. The identical image

data processing during encryption reveals a similarity between setups while the main distinctions stem from CPU execution differences and processing timings.

Table 4.8 Comparative Analysis of Grayscale Image Metrics in AWS and VS code

Metrics	**Encryption time**	**Decryption time**	**CPU usage during encryption**	**CPU usage during decryption**	**Peak memory during encryption**	**Peak memory during decryption**
VS Code	0.0109 seconds	0.0836 seconds	3.60 %	5.40 %	0.036 MB	7.6514 MB
Amazon AWS Linux	0.102 seconds	0.0494 seconds	100.00%	100.00%	0.0340 MB	7.6480 MB

5. Conclusion

The project implements a secure visual cryptography image sharing system by using XOR encryption method, Modular Arithmetic (MA), CMYK halftone processing, Pixel Expansion together with Bit Level Decomposition (BLD). Under the XOR method image encryption/decryption completed at the fastest rate of all approaches in 0.0109 seconds using limited CPU capacity alongside nominal memory requirements thereby making XOR appropriate for lightweight image applications. The combination of secure CMYK halftone processing methods with efficient execution time allowed handling of coloured image content effectively. The performance metrics of MA reached equilibrium by using medium memory allocations between 143 MB and 159 MB. The binary image processing based on Pixel Expansion produced successful results by giving overlapping images and extracted images with different reconstructed metrics that showed independent metric profiles. The encryption and decryption process in BLD method used a significant amount of computational resources and ran for more than 74 seconds with memory usage exceeding others thus limiting its suitability for real-time applications. The system used PSNR and SSIM metrics to evaluate imaging quality alongside hash matching and NCORR to evaluate security aspects for distortion. The testing on the AWS EC2 platform using S3 demonstrated the system's scalability across domains of forensics, defence, and healthcare because it enables secure and efficient image transport.

6. Future Work and Plan

The future system development will integrate flexible encryption algorithms able to select between XOR and either MA or CMYK encryption based on incoming image resolutions and sensitivity values. Security systems become stronger through machine learning-based detection of security vulnerabilities that simultaneously provides recommended secure ways to distribute shares. The development of the system requires additional features which incorporate temporal visual cryptography encryption for safeguarding surveillance videos and medical video distribution.

The robustness of the system can be enhanced through multi-factor authentication alongside blockchain share tracking that guarantees both share integrity and traceability during cloud transmission process. The system's real-time performance can be made more efficient when developed through parallel processing approaches utilizing CUDA for Graphical Processing Unit acceleration. A friendly web interface alongside a mobile application can be developed to ensure healthcare and defence and forensic institution workers without technical expertise can access the tool easily.

The future development of this project can also concentrate on improving security features and usability aspects of the visual cryptography system. The research method will require expansion toward handling dynamic media contents such as video sequences to establish secure temporal visual cryptography for video communications.

6.1 Progress Report Gantt Schedule Chart

1)Designing Phase

I started the project by creating a blueprint for system architecture along with selecting the cryptographic methods which included XOR, MA, BLD, CMYK Decomposition and PE. I established functional specifications for image processing with three different image types while defining performance parameters which consisted of encryption/decryption speed alongside CPU and RAM consumption together with PSNR and SSIM and Normalized Correlation. The next step involved my selection of AWS EC2 for application hosting alongside AWS S3 for encrypted share storage and retrieval.

2)Development Phase

The fundamental operations of the system were executed in the development phase. The development involved my creation of discrete modules which handled encryption and decryption operations by implementing chosen cryptographic algorithms. The system includes automated performance metric calculations which operate in real-time. The system employs logic to select input image types with help of GUI so it can direct the images to suitable encryption and decryption processing paths. The system afterwards integrates AWS services to offer effortless image and share uploading and downloading

followed by encryption process and evaluating results at the final stage. I conducted extensive testing of encryption modules to ensure their strength and flexibility during this phase.

3)Testing Phase

The testing phase required evaluation of the system through multiple grayscale images, binary images and coloured images. The testing included measuring both encryption time and decryption time duration with real-time assessments of CPU consumption and memory utilization together with share reconstruction verification. The visual quality assessment of decrypted images depended on PSNR ,NCORR and SSIM calculations while hash-based methods confirmed data integrity verification. The assessment focused on understanding which cryptographic methods managed different image formats while maintaining successful original image recovery from encrypted shares.

4)Evaluation of Methods Phase

The system performance analysis included theoretical concepts as well as observed outcome measurements. XOR-based encryption offered the fastest encryption speed together with minimal resource utilisation but BLD provided the good visual performance despite using significant system resources. The analysis focused on PE's ability to produce two output image types along with its adjustable display features. The cloud functionality proved reliable through successful retrieval of AWS S3 encrypted shares without any data damage or corruption.

5)Reporting Phase

I assembled and finalized all project documentation by presenting system design alongside methodology descriptions along with implementation results and assessments of performance in the final stage. The project required creation of visual elements which included bar graphs for theoretical analysis and implemented plots for visualizing original and decrypted images alongside their difference heatmap. The documentation includes my recommendations for future improvements which include artificial intelligence for path selection. A complete final evaluation included a presentation with demonstration materials based on my work.

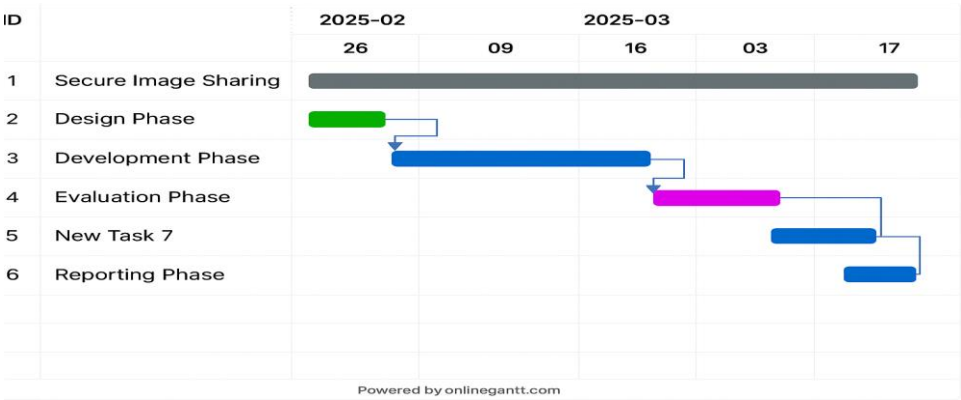


Figure 6.1 Gantt Chart

PROGESS REPORT

Week	Milestone / Activity	Status
1	Focusing and finding a relevant problem, related to data sharing.	Completed
2	Research of Visual Cryptography and methods for sharing protected images.	Completed
3	Did analysis to examine current secure image sharing solutions and to find potential enhancement opportunities.	Completed
4	The project setup stage was initiated with the development of the structure of the project design.	Completed
5	Familiarizing with methods like XOR, PE, BLD, MA and CMYK decomposition method which are to be used.	Completed
6	Presentation making of methods to be used and including a research table to review existing schemes and their limitations	Completed
Break	TA-1 Exams	Academic Break
7	Designing algorithms for encryption and decryption using the logic of selected methods for project and generating shares ranging from 2-8 for a single grayscale, binary or coloured image.	Completed
8	Now enhancing the scope of the project by taking 3 input images instead of one.	Academic Break
9	Selection of metrics like PSNR, SSIM, NCORR and Hash verification for comparison of original and reconstructed image.	Completed
10	Using tkinter GUI to select input images according to the wish of the user.	Completed
11	Solving errors which were generated whilst evaluating the images.	Completed
12	Mid-Sem Exam Week (buffer and prep included)	Academic Break
13	Created channels of server – client architecture containing 1 server and 3 clients for distributing shares of selected input image and writing code for client 3 to combine shares of client 1 and client 2 and at last visually comparing original and decrypted images.	Completed
14	Mid-Project Viva/ TA-2	Completed
15	Produced graphical representations to examine how methods performed their encryption-decryption time periods alongside their CPU usage levels during	Completed

	encryption-decryption operations as well as their memory usage during encryption-decryption processes.	
16	The same procedure was executed on AWS Amazon Linux instances through VS Code for the purpose of comparison between platform execution efficiency.	Completed
17	The project moved into documentation alongside report composition.	Completed
Break	End-Sem Preparation	Study Break
18	End Semester Exams	Academic Break
19	The remaining tasks involved terminal edits for texts and finalizing paper presentation before establishing appendices.	Completed
20	Final submission of report and deliverables	In Progress

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